PROJECT AND TEAM INFORMATION

Project Title

EduTrack: Intelligent Student Performance and Academic Record Management System

Student / Team Information

Team Name: ***Team ID:***	**EduTrack** **DSCPP-III-2026-T287**
Team member 1 (Team Lead) **Mohammad Rayyan Farooqui** Student ID: 2510370176 2510370176@geu.ac.in	
Team member 2 **Shiv Tyagi** Student ID: 2510370194 2510370194@geu.ac.in	
Team member 3 **Bhaskar Bahukhandi** Student ID: 2510370009 2510370009@geu.ac.in	
Team member 4 **Aditya Kukrety** Student ID: 2510012343 2510012343@geu.ac.in	

Mentor	Mr. Abhinav Kotnala
Email	abhinavkotnala.cse@geu.ac.in

PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

When the number of students becomes large, keeping track of student details and academic records can become difficult. Information such as names, Roll Numbers, subject marks, grades and other academic details may be stored separately, which can make it harder to find or update the required information.EduTrack is proposed as a C++ program that can keep these records in one place and make common operations easier to perform.

The program will store student details along with subject-wise marks and will calculate the total marks, percentage and grade automatically.The system will also include searching and sorting so that records can be found and arranged when needed.The main idea of the project is not only to store student information but also to use the stored data for simple academic analysis. For example, the program can find the student with the highest marks, prepare a rank list and show how an individual student's performance compares with the class averageThe project will also give us a chance to use the concepts covered in Data Structures and C++, such as vectors or dynamic arrays, searching, sorting, queues, stacks, Object-Oriented Programming and file handling.Overall, EduTrack is meant to be a practical way of applying the concepts learned in the course to a student record management problem.

State of the Art / Current solution (1 pt)

Student records are usually maintained using spreadsheets, separate files, manual records or student management systems. These methods can store the required information, but some tasks still have to be done manually.For example, when the number of students increases, finding a particular student, changing their details, calculating percentages, arranging students according to marks and preparing a rank list can take more time.A basic Student Management System can handle common operations such as adding, searching, updating and deleting student records. However, it may not include much academic analysis along with these operations.EduTrack is planned to combine both record management and academic processing in one program. Students can be searched using their Roll Number, and sorting can be used to arrange them according to marks or percentage.
The sorted records can then be used to create a rank list and find the top-performing student. The system will also include features such as performance categories, subject-wise performance and comparison of a student's marks with the class average.
In this way, the project will bring together C++, Object-Oriented Programming and Data Structures and Algorithms in a single student academic record system.

Project Goals and Milestones (2 pts)

The primary goal of EduTrack is to develop an efficient C++-based student academic record management system that combines student record management with academic processing, searching, sorting, ranking and performance analysis.
The major goals of the project are:

1. Develop a C++-based student academic record management system.
2. Store student details and subject-wise marks efficiently.
3. Automatically calculate total marks, percentage and grades.
4. Provide searching functionality for student records.
5. Allow student records to be updated and deleted.
6. Apply suitable sorting techniques to organize students according to marks or percentage.
7. Generate a dynamic academic rank list.
8. Identify the top-performing student.
9. Categorize students according to their academic performance.
10. Display subject-wise performance of students.
11. Compare individual student performance with the class average.
12. Maintain permanent student records using C++ file handling.
13. Demonstrate meaningful integration of Object-Oriented Programming and Data Structures and Algorithms.

Initial Milestones

1. Problem identification and formulation.
2. Background research and requirement analysis.
3. Identification of major system features.
4. Design of the proposed system workflow.
5. Mapping of C++ OOP and DSA concepts to individual system modules.
6. Development of the basic C++ implementation.
7. Implementation of student record operations.
8. Integration of searching and sorting algorithms.
9. Implementation of academic performance analysis.
10. Integration of file handling for permanent storage.
11. Testing and debugging.
12. Final evaluation and refinement.

Project Approach (3 pts)

The proposed EduTrack system will be developed as a modular C++ application. The development will start with basic student information such as Roll Number, Name, Branch and Semester, along with marks obtained in different subjects.

A Student class will be created to store the information of each student. Basic Object-Oriented Programming concepts will be used to keep the data and functions properly organized. Encapsulation will be used so that student information can be accessed in a controlled way. Constructors can also be used to initialize the student objects when they are created.

Student records will be stored using suitable data structures such as vectors or dynamic arrays. This will make it easier to add and manage multiple student records while the program is running.

The system will calculate total marks, percentage and grades for every student. Searching methods will be used to find student records. Linear Search can be used for normal searching, while Binary Search can be used when the records are already sorted.

Sorting algorithms will arrange students according to their marks or percentage. The sorted records can then be used to create a rank list and find the top-performing student.

The system will also include basic academic analysis, such as subject-wise performance, student performance categories and comparison with the class average.

Queue and Stack concepts may also be used where needed. A Queue can keep students waiting for processing, while a Stack can store recently completed operations.

File handling using C++ and the fstream library will be used for permanent storage. This will allow records to be saved in files and loaded again when the program is run.

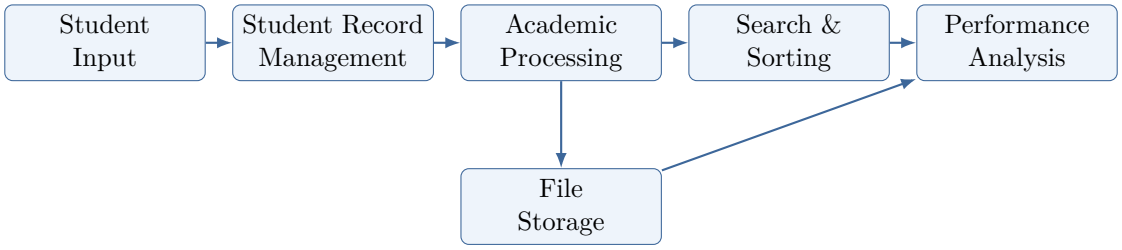
- C++ programming language.
- Object-Oriented Programming.
- Classes and Objects.
- Encapsulation.
- Constructors.
- Inheritance and Polymorphism where required.
- Vector / Dynamic Array.
- Queue.
- Stack.
- Linear Search.
- Binary Search.
- Sorting Algorithms.
- Algorithmic processing.
- C++ File Handling using fstream.

Development Methodology

The development will follow the stages of problem identification, background research, requirement analysis, system design, DSA and OOP mapping, implementation, testing, debugging and evaluation.

System Architecture (High Level Diagram)(2 pts)

The EduTrack system will work through a number of connected modules. First, the student enters basic information such as Roll Number, Name, Branch, Semester and subject-wise marks through the Student Input module.This information is then passed to the Student Record Management module, where student records can be added, displayed, searched, updated or deleted.The Academic Processing module will work on the marks entered for each student. It will calculate the total marks, percentage and grade for each student. The Search and Sorting section will find student records and arrange them by marks or percentage. The sorted records will be used to prepare the rank list and find the top-performing student.
The Performance Analysis module will also compare a student's result with the class average and show subject-wise performance.The File Storage module will save records in files so they can be accessed later.



System Components

- Student Record Management: Adds, displays, searches, updates and deletes student records.
- Academic Processing:Helps in Calculating total marks, percentage and grades.
- Search and Sorting: Find the records and sorts students by marks or percentage.
- Performance Analysis: Prepares the rank list, finds the top-performing student and compares results with the class average marks.
- File Storage: Saves the records using C++ file handling.

Project Outcome / Deliverables (1 pts)

EduTrack will provide the following:

1. A working C++ program for student record management system.
2. Adding, displaying, searching, updating and deleting the records.
3. Automatic calculation of marks, percentage and grades.
4. Sorting the students by marks or percentage.
5. A rank list based on student performance.
6. Identification of the top-performing student.Academic performance categorization.
7. Subject-wise student performance information.
8. Comparison of individual student performance with the class average.
9. Permanent storage and retrieval of student records using C++ file handling.
10. Practical demonstration of C++ Object-Oriented Programming.
11. Practical application of Data Structures and Algorithms to a real-world academic management problem.

The final project is expected to demonstrate that DSA concepts can be used not only for theoretical problems but also for organizing, searching, processing and analyzing real-world academic information.

Assumptions

The proposed system assumes that valid student information and subject-wise marks will be provided as input.
Each student is assumed to have a unique Roll Number that can be used for identification and searching.
The grading and percentage calculation rules used by the system are assumed to be defined before implementation.
It is assumed that users entering or modifying student records will provide information in the expected format.
The current project is primarily intended as an academic project for demonstrating C++ Object-Oriented Programming and Data Structures and Algorithms. Advanced institutional database integration, authentication and large-scale deployment are outside the current proposed scope.

References

The following categories of resources will be used during the development and documentation of the EduTrack project:

1. C++ programming and Object-Oriented Programming reference materials.
2. Data Structures and Algorithms reference materials.
3. C++ Standard Library documentation.
4. Documentation related to C++ vectors and dynamic arrays.
5. Documentation and reference material for searching and sorting algorithms.
6. C++ file handling and fstream documentation.
7. Academic management system reference material consulted during the problem analysis and proposal development.
8. Course material related to Data Structures and C++.

Technical References

- C++ Standard Library documentation.
- Data Structures and Algorithms course resources.
- C++ Object-Oriented Programming reference resources.
- File handling and fstream documentation.